Safouene Ziadi

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Final-year software engineering student focused on AI Systems , Deep Learning, and Interpretable Machine Learning. Experienced in building neural architectures from scratch using PyTorch and developing AI systems involving multi-agent workflows and computer vision. Seeking AI engineering or research internships/jobs.

## Education

**Higher Institute of Informatics and Mathematics** — Monastir
Diploma in Software Engineering — 2024 - 2027

**Higher Institute of Informatics and Mathematics** — Monastir
Software Pre-Engineering Program — 2022-2024

## Experience

**Nachd-IT** | Sousse | 2025-2026

- **AI Engineer** : Developing an AI-powered chatbot capable of interacting with databases through natural language queries and performing secure SQL-based operations dynamically.
- **AI Research Intern** : Conducted research on the integration of artificial intelligence within educational platforms, exploring AI-driven approaches to enhance learning experiences and student support systems.

**We Are Technology Center** | Monastir | 2025-2026

- **Machine Learning & Deep Learning Instructor** : Delivered 50+ hands-on training sessions on machine learning and deep learning concepts, guiding students through practical AI project development using Python and PyTorch.
- **C/C++ Instructor** : Taught programming fundamentals, problem-solving, and algorithmic thinking through structured C and C++ training sessions for 20+ Students.

## Projects

**P-SPINE Architecture (Research Project) | PyTorch, NLP | Github**
- Designed a novel neural architecture focused on improving the interpretability of word embeddings while preserving semantic representation quality
- Explored methods for making latent embedding dimensions more explainable and human-interpretable
- Conducted independent research and experimentation, with the project being developed as a personal research paper

**QARAI Agent Guard | AI Safety , Agentic AI , Langchain | Github**
- Built Qarai Agent Guard, a Python toolkit using ML and pattern-based detection to catch prompt injection, jailbreaks, and PII leaks in AI agents
- Designed severity-based policies (block, redact, warn, quarantine) with multilingual detection support
- Integrated it into LangChain and CrewAI via middleware and hooks to enforce security across agent I/O

**Doctor Copilot | CrewAI , Multi-Agent AI, Python | Github**
- Built an AI-powered multi-agent system designed to assist mental health professionals by analyzing patient digital behavior patterns
- Developed collaborative AI agents capable of processing, summarizing, and generating behavioral insights from digital interactions
- Focused on creating an extensible architecture for intelligent healthcare support and decision assistance

**Vision Transformer from Scratch | PyTorch, Deep Learning | Github**
- Implemented a Vision Transformer (ViT) architecture entirely from scratch using PyTorch without relying on pretrained components
- Built core transformer components including patch embeddings, positional encoding, multi-head self-attention, and encoder blocks
- Trained and evaluated the model on the MNIST dataset, achieving over 90% classification accuracy

## Skills

**Programming Languages** : Python ,C , C++ , Java , Javascript , SQL , PHP

**Frameworks and Libraries** : Pytorch , Keras/Tensorflow , Langchain , CrewAI , React , FastApi , Laravel

**Tools** : Docker , Git/Github , DBeaver , Mysql , Postman

## Volunteer Experience

- **Vice President & Event Organizer (CPU-ISIMM CLUB)** : Coordinated technical events, managed team activities, and contributed to organizing student-focused workshops and initiatives.
- **Head of AI Committee (CPU-ISIMM CLUB)** : Led the AI committee by organizing and delivering multiple training sessions for 100+ students on machine learning, deep learning, and AI development topics.